

Model Development Phase Template

Date	11 July 2024
Team ID	SWTID1720011518
Project Title	WCE curated Colon Disease Classification using Deep Learning
Maximum Marks	10 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code will be showcased in the future through a screenshot. The model validation and evaluation report will include a summary and training and validation performance metrics for multiple models, presented through respective screenshots.

Initial Model Training Code (5 marks):

VGG16:

```
[ ] for layer in vgg.layers:
    layer.trainable = False

[ ] x = Flatten()(vgg.output)

[ ] output = Dense(4,activation = "softmax")(x)

[ ] vgg16 = Model(vgg.input , output)

[ ] vgg16.summary()

Model: "model"
-----
Layer (type)                 Output Shape              Param #
-----
input_1 (InputLayer)         [(None, 224, 224, 3)]     0
block1_conv1 (Conv2D)        (None, 224, 224, 64)      1792
block1_conv2 (Conv2D)        (None, 224, 224, 64)      36928
```

```
[ ] vgg16.compile(optimizer = "adam",loss = "categorical_crossentropy",metrics = ["accuracy"])

history = vgg16.fit(train , validation_data= test,epochs = 5)

Epoch 1/5
160/160 [=====] - 107s 622ms/step - loss: 0.2567 - accuracy: 0.9134 - val_loss: 0.1309 - val_accuracy: 0.9469
Epoch 2/5
160/160 [=====] - 100s 624ms/step - loss: 0.0870 - accuracy: 0.9688 - val_loss: 0.0259 - val_accuracy: 0.9912
Epoch 3/5
160/160 [=====] - 100s 628ms/step - loss: 0.0551 - accuracy: 0.9806 - val_loss: 0.1200 - val_accuracy: 0.9481
Epoch 4/5
160/160 [=====] - 82s 514ms/step - loss: 0.0398 - accuracy: 0.9847 - val_loss: 0.1132 - val_accuracy: 0.9522
Epoch 5/5
160/160 [=====] - 81s 509ms/step - loss: 0.0272 - accuracy: 0.9909 - val_loss: 0.0190 - val_accuracy: 0.9937
```

Resnet50:

```
for layer in resnet50.layers:
    layer.trainable = False

x1 = Flatten()(resnet50.output)

output = Dense(4,activation = "softmax")(x1)

resnet = Model(resnet50.input , output)

resnet.summary()
```

Layer (Batch Normalization)	Output Shape	Number of Parameters	Connected Layer
conv5_block2_2_bn (Batch Normalization)	(None, 7, 7, 512)	2048	['conv5_block2_2_conv[0][0]']
conv5_block2_2_relu (Activation)	(None, 7, 7, 512)	0	['conv5_block2_2_bn[0][0]']
conv5_block2_3_conv (Conv2D)	(None, 7, 7, 2048)	1050624	['conv5_block2_2_relu[0][0]']
conv5_block2_3_bn (Batch Normalization)	(None, 7, 7, 2048)	8192	['conv5_block2_3_conv[0][0]']

```
resnet.compile(loss = "categorical_crossentropy", optimizer = "adam", metrics = ["accuracy"])

history3 = resnet.fit(train , validation_data = test , epochs = 5)
```

Epoch	Loss	Accuracy	Val Loss	Val Accuracy
Epoch 1/5	0.4264	0.8394	0.3856	0.8422
Epoch 2/5	0.5056	0.8156	0.3810	0.8481
Epoch 3/5	0.5100	0.8231	0.3660	0.8537
Epoch 4/5	0.3765	0.8594	0.2215	0.9191
Epoch 5/5	0.3833	0.8625	0.2326	0.9047

InceptionV3:

```
for layer in inception_v3.layers:
    layer.trainable = False

x3 = Flatten()(inception_v3.output)

output3 = Dense(4,activation = "softmax")(x3)

inception = Model(inception_v3.input, output3)

inception.summary()
```

Layer (Conv2D)	Output Shape	Number of Parameters	Connected Layer
conv2d_88 (Conv2D)	(None, 8, 8, 384)	442368	['activation_86[0][0]']
conv2d_91 (Conv2D)	(None, 8, 8, 384)	442368	['activation_90[0][0]']
conv2d_92 (Conv2D)	(None, 8, 8, 384)	442368	['activation_90[0][0]']
average_pooling2d_8 (Average Pooling2D)	(None, 8, 8, 2048)	0	['mixed9[0][0]']
conv2d_85 (Conv2D)	(None, 8, 8, 320)	655360	['mixed9[0][0]']
batch_normalization_87 (Batch Normalization)	(None, 8, 8, 384)	1152	['conv2d_87[0][0]']

```
inception.compile(loss = "categorical_crossentropy", optimizer = "adam", metrics = ["accuracy"])

history4 = inception.fit(train, validation_data = test, epochs = 5)
```

Epoch	Loss	Accuracy	Val Loss	Val Accuracy
Epoch 1/5	2.6529	0.8931	0.8691	0.9375
Epoch 2/5	0.5101	0.9691	0.6682	0.9525
Epoch 3/5	0.3904	0.9747	1.3684	0.9362
Epoch 4/5	0.4131	0.9778	0.6467	0.9675
Epoch 5/5	0.2585	0.9850	4.9814	0.8537

Model Validation and Evaluation Report (5 marks):

Model	Summary	Training and Validation Performance Metrics
VGG16	<pre> vgg.summary() Model: "vgg16" Layer (type) Output Shape Param # ----- input_1 (InputLayer) [None, 224, 224, 3] 0 block1_conv1 (Conv2D) [None, 224, 224, 64] 1792 block1_conv2 (Conv2D) [None, 224, 224, 64] 36928 block1_pool (MaxPooling2D) [None, 112, 112, 64] 0 block2_conv1 (Conv2D) [None, 112, 112, 128] 73856 block2_conv2 (Conv2D) [None, 112, 112, 128] 147584 block2_pool (MaxPooling2D) [None, 56, 56, 128] 0 block3_conv1 (Conv2D) [None, 56, 56, 256] 295168 block3_conv2 (Conv2D) [None, 56, 56, 256] 590080 block3_conv3 (Conv2D) [None, 56, 56, 256] 590080 block4_pool (MaxPooling2D) [None, 14, 14, 512] 0 block5_conv1 (Conv2D) [None, 14, 14, 512] 2359808 block5_conv2 (Conv2D) [None, 14, 14, 512] 2359808 block5_conv3 (Conv2D) [None, 14, 14, 512] 2359808 block5_pool (MaxPooling2D) [None, 7, 7, 512] 0 flatten (Flatten) [None, 25088] 0 dense (Dense) [None, 4] 100356 Total params: 14815044 (56.51 MB) Trainable params: 100356 (392.02 KB) Non-trainable params: 14714688 (56.13 MB) </pre>	<pre> History = vgg16.fit(train, validation_data=test, epochs = 5) Epoch 1/5: - 107s 628ms/step - loss: 0.2567 - accuracy: 0.9134 - val_loss: 0.1389 - val_accuracy: 0.9469 Epoch 2/5: - 100s 624ms/step - loss: 0.0870 - accuracy: 0.9688 - val_loss: 0.0259 - val_accuracy: 0.9912 Epoch 3/5: - 100s 628ms/step - loss: 0.0751 - accuracy: 0.9806 - val_loss: 0.1200 - val_accuracy: 0.9481 Epoch 4/5: - 82s 114ms/step - loss: 0.0398 - accuracy: 0.9847 - val_loss: 0.1112 - val_accuracy: 0.9522 Epoch 5/5: - 81s 109ms/step - loss: 0.0272 - accuracy: 0.9909 - val_loss: 0.0100 - val_accuracy: 0.9917 </pre>
Resnet50	<pre> resnet.summary() conv5_block2_2_bn (BatchNorm (None, 7, 7, 512) 2048 ['conv5_block2_2_conv[0][0]'] rmulization) conv5_block2_2_relu (Activ (None, 7, 7, 512) 0 ['conv5_block2_2_bn[0][0]'] ation) conv5_block2_3_conv (Conv2 (None, 7, 7, 2048) 1058624 ['conv5_block2_2_relu[0][0]'] D) conv5_block2_3_bn (BatchNorm (None, 7, 7, 2048) 8192 ['conv5_block2_3_conv[0][0]'] rmulization) conv5_block2_3_add (Add) (None, 7, 7, 2048) 0 ['conv5_block2_out[0][0]', 'conv5_block2_3_bn[0][0]'] conv5_block2_out (Activati (None, 7, 7, 2048) 0 ['conv5_block2_add[0][0]'] on) conv5_block3_1_conv (Conv2 (None, 7, 7, 512) 1049088 ['conv5_block2_out[0][0]'] D) conv5_block3_1_bn (BatchNorm (None, 7, 7, 512) 2048 ['conv5_block3_1_conv[0][0]'] rmulization) conv5_block3_1_relu (Activ (None, 7, 7, 512) 0 ['conv5_block3_1_bn[0][0]'] ation) conv5_block3_3_bn (BatchNorm (None, 7, 7, 2048) 8192 ['conv5_block3_3_conv[0][0]'] rmulization) conv5_block3_3_add (Add) (None, 7, 7, 2048) 0 ['conv5_block2_out[0][0]', 'conv5_block3_3_bn[0][0]'] conv5_block3_out (Activati (None, 7, 7, 2048) 0 ['conv5_block3_add[0][0]'] on) flatten_1 (Flatten) (None, 100352) 0 ['conv5_block3_out[0][0]'] dense_1 (Dense) (None, 4) 401412 ['flatten_1[0][0]'] Total params: 23989124 (91.51 MB) Trainable params: 401412 (1.53 MB) Non-trainable params: 23587712 (89.98 MB) </pre>	<pre> History = resnet.fit(train, validation_data = test, epochs = 5) Epoch 1/5: - 81s 509ms/step - loss: 0.4204 - accuracy: 0.0394 - val_loss: 0.3056 - val_accuracy: 0.0422 Epoch 2/5: - 81s 504ms/step - loss: 0.5456 - accuracy: 0.0156 - val_loss: 0.3810 - val_accuracy: 0.0481 Epoch 3/5: - 80s 500ms/step - loss: 0.5100 - accuracy: 0.0231 - val_loss: 0.3600 - val_accuracy: 0.0537 Epoch 4/5: - 81s 506ms/step - loss: 0.3705 - accuracy: 0.0394 - val_loss: 0.2215 - val_accuracy: 0.0193 Epoch 5/5: - 80s 500ms/step - loss: 0.3833 - accuracy: 0.0625 - val_loss: 0.2120 - val_accuracy: 0.0047 </pre>

Inception V3

```
inception.summary()
batch_normalization_87 (Ba (None, 8, 8, 384)
tchNormalization) 1152 ["conv2d_87[0][0]"]
batch_normalization_88 (Ba (None, 8, 8, 384)
tchNormalization) 1152 ["conv2d_88[0][0]"]
batch_normalization_91 (Ba (None, 8, 8, 384)
tchNormalization) 1152 ["conv2d_91[0][0]"]
batch_normalization_92 (Ba (None, 8, 8, 384)
tchNormalization) 1152 ["conv2d_92[0][0]"]
conv2d_93 (conv2D) (None, 8, 8, 192) 393216 ["average_pooling2d_8[0][0]"]
batch_normalization_85 (Ba (None, 8, 8, 320)
tchNormalization) 960 ["conv2d_85[0][0]"]
activation_87 (Activation) (None, 8, 8, 384) 0 ["batch_normalization_87[0][0]"]
activation_88 (Activation) (None, 8, 8, 384) 0 ["batch_normalization_88[0][0]"]

activation_93 (Activation) (None, 8, 8, 192) 0 ["batch_normalization_93[0][0]"]
mixed10 (Concatenate) (None, 8, 8, 2048) 0 ["activation_85[0][0]",
"mixed9_1[0][0]",
"concatenate_1[0][0]",
"activation_93[0][0]"]
flatten_2 (Flatten) (None, 131072) 0 ["mixed10[0][0]"]
dense_4 (dense) (None, 4) 524292 ["flatten_2[0][0]"]

Total params: 22327876 (85.17 MB)
Trainable params: 524292 (2.00 MB)
Non-trainable params: 21803584 (83.17 MB)
```

```
history4 = inception.fit(train, validation_data = test, epochs = 5)

Epoch 1/5
100/100 [=====] - loss: 2.6529 - accuracy: 0.8091 - val_loss: 0.8091 - val_accuracy: 0.9312
Epoch 2/5
100/100 [=====] - loss: 0.5181 - accuracy: 0.9691 - val_loss: 0.6682 - val_accuracy: 0.9525
Epoch 3/5
100/100 [=====] - loss: 0.3984 - accuracy: 0.9747 - val_loss: 1.3684 - val_accuracy: 0.9362
Epoch 4/5
100/100 [=====] - loss: 0.4131 - accuracy: 0.9778 - val_loss: 0.6467 - val_accuracy: 0.9675
Epoch 5/5
100/100 [=====] - loss: 0.2585 - accuracy: 0.9858 - val_loss: 4.9814 - val_accuracy: 0.8537
```

Model Development Phase Template

Date	11 July 2024
Team ID	SWTID1720011518
Project Title	WCE curated Colon Disease Classification using Deep Learning
Maximum Marks	5 Marks

Model Selection Report

In the model selection report for future deep learning and computer vision projects, various architectures, such as CNNs or RNNs, will be evaluated. Factors such as performance, complexity, and computational requirements will be considered to determine the most suitable model for the task at hand.

Model Selection Report:

Model	Description
VGG16	By using feature extraction, we have used the VGG16 pre-trained model. One Dense and one Flatten layer is used. The loss function is categorical_crossentropy, optimizer is “adam” and metrics for evaluation is accuracy. For 5 epochs: The Training Loss is: 0.0272 The Training Accuracy is: 0.9909 The Validation Loss is: 0.0190 The Validation Accuracy is: 0.9907

Resnet50	By using feature extraction, we have used the Resnet50 pre-trained model. One Dense and one Flatten layer is used. The loss function is categorical_crossentropy, optimizer is “adam” and metrics for evaluation is accuracy. For 5 epochs: The Training Loss is: 0.3833 The Training Accuracy is: 0.8625 The Validation Loss is: 0.2326 The Validation Accuracy is: 0.9047
InceptionV3	By using feature extraction, we have used the InceptionV3 pre-trained model. One Dense and one Flatten layer is used. The loss function is categorical_crossentropy, optimizer is “adam” and metrics for evaluation is accuracy. For 5 epochs: The Training Loss is: 0.2585 The Training Accuracy is: 0.9850 The Validation Loss is: 4.9814 The Validation Accuracy is: 0.8537